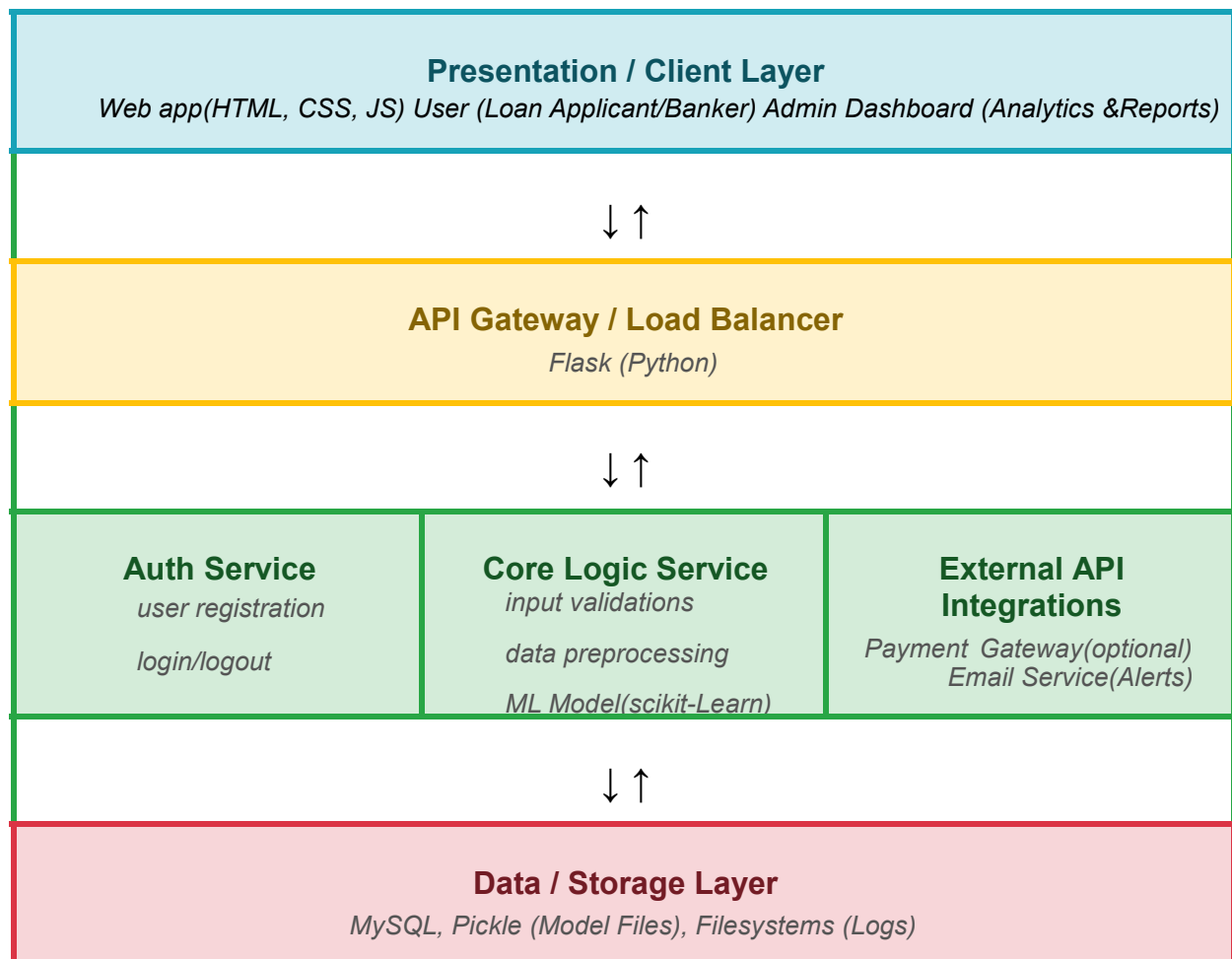


Solution Architecture

Date	
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Smart Lender Loan Prediction System follows a layered architecture consisting of the Presentation Layer, API Gateway, Core Logic Service, and Data Storage Layer. The system collects applicant information through a web interface, processes the data using machine learning algorithms, and predicts loan eligibility accurately and efficiently. This architecture ensures scalability, reliability, and faster loan approval decisions.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface for applicants to enter loan details and view prediction results.	HTML, CSS, JavaScript, Bootstrap
API Gateway	Receives user requests, processes form data, and connects the frontend with the machine learning model.	Flask (Python)
Microservices	Performs data preprocessing, executes the trained machine learning model, and predicts loan eligibility.	Python, scikit-learn, pandas, numpy, joblib
Database	Stores datasets, trained model files, application data, and prediction results.	CSV Files, Pickle (.pkl) Model File, File System